

Support Vector Machine: Comprehensive Explanation and Implementation Guide

1 Understanding the Classification Problem

- We have a dataset consisting of m samples: $\{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$, where each $x_i \in \mathbb{R}^n$ is a feature vector, and $y_i \in \{-1, +1\}$ is the corresponding class label.
- The goal is to find a decision boundary (hyperplane) that separates the data points of different classes.
- The objective of the SVM algorithm is to find the optimal hyperplane that maximizes the margin between the classes.

2 Defining the Hyperplane

- The hyperplane is defined by the equation:

Hyperplane Equation

$$w \cdot x + b = 0$$

- Here, w is the weight vector, and b is the bias term.
- The classification decision for any data point x_i is based on the sign of $w \cdot x_i + b$:

Classification Decision

If $y_i(w \cdot x_i + b) \geq 1$, then x_i is correctly classified.

3 The Optimization Problem

- To find the values of w and b that minimize the norm of the weight vector $\|w\|$, we solve:

Optimization Problem

$$\min_{w, b} \frac{1}{2} \|w\|^2$$

subject to:

$$y_i(w \cdot x_i + b) \geq 1 \quad \forall i$$

- Minimizing $\|w\|$ is equivalent to maximizing the margin between the hyperplane and the nearest data points.

4 Lagrangian and Dual Formulation

- The optimization problem is challenging to solve directly due to constraints. To simplify, we introduce Lagrange multipliers $\alpha_i \geq 0$ for each constraint:

Lagrangian

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{i=1}^m \alpha_i [y_i (w \cdot x_i + b) - 1]$$

- The dual problem is derived by maximizing the Lagrangian with respect to the multipliers α_i :

Dual Problem

$$\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j y_i y_j (x_i \cdot x_j)$$

subject to:

$$\sum_{i=1}^m \alpha_i y_i = 0 \quad \text{and} \quad \alpha_i \geq 0$$

5 Solving the Dual Problem

- After solving the dual problem, the optimal values of α_i are used to compute the weight vector w :

Weight Vector

$$w = \sum_{i=1}^m \alpha_i y_i x_i$$

- The bias b can be computed from any support vector x_k (where $\alpha_k > 0$):

Bias

$$b = y_k - w \cdot x_k$$

6 Soft Margin for Non-Separable Data

- Real-world data is often not perfectly separable. To handle this, SVM introduces slack variables ξ_i to allow some misclassifications:

Slack Variables

$$y_i (w \cdot x_i + b) \geq 1 - \xi_i \quad \forall i$$

- The updated objective function is:

Updated Objective Function

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^m \xi_i$$

- Here, C is a regularization parameter that controls the trade-off between maximizing the margin and minimizing the classification error.

7 Kernel Trick for Non-Linear Separation

- For non-linearly separable data, SVM uses a kernel function $K(x_i, x_j)$ to map the input data into a higher-dimensional space:

Kernel Function

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

- Common kernel functions include:
 - Linear Kernel: $K(x_i, x_j) = x_i \cdot x_j$
 - Polynomial Kernel: $K(x_i, x_j) = (x_i \cdot x_j + 1)^d$
 - Radial Basis Function (RBF) Kernel: $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$

8 Final Classification

- The decision function for classifying a new data point x is:

Decision Function

$$f(x) = \text{sign} \left(\sum_{i=1}^m \alpha_i y_i K(x_i, x) + b \right)$$